



NABAPRAVAT ACADEMY FOR MANAGEMENT EDUCATION

Cuttack, Odisha

Diabetes Prediction System using Machine Learning

Industrial Internship Report – Data Science & Machine
Learning (upSkill Campus & UCT)

Prepared by:

Soumya Priyadarshinee Sahoo

Academic Year 2024–25

Submitted in partial fulfillment of the requirements for the
Industrial Internship Program • 2024–25



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NABAPRAVAT ACADEMY FOR MANAGEMENT EDUCATION

Affiliated to Utkal University

Cuttack, Odisha

Motto: DISCIPLINE ? DEDICATION ? DETERMINATION

INDUSTRIAL INTERNSHIP REPORT

On

DIABETES PREDICTION SYSTEM USING MACHINE LEARNING

Domain: Data Science and Machine Learning

Submitted in partial fulfillment of Industrial Internship

Submitted By:

Soumya Priyadarshinee Sahoo

BCA / MCA - Data Science Batch

Roll No: _____

Under Guidance Of:

upSkill Campus & The IoT Academy

in collaboration with

UniConverge Technologies Pvt Ltd (UCT)

Academic Year: 2024-2025



Page 2 / 16 - CERTIFICATE

CERTIFICATE

This is to certify that Ms. Soumya Priyadarshinee Sahoo, student of Nabapravat Academy for Management Education, Cuttack has successfully completed 6 weeks Industrial Internship in Data Science and Machine Learning domain organized by upSkill Campus and The IoT Academy in collaboration with UniConverge Technologies Pvt Ltd.

Project Title: Diabetes Prediction System using Machine Learning

Internship ID: USC/UCT/AUG26/2024/XXXX

Duration: 6 Weeks

Domain: Data Science & ML

During this period, she has completed all modules, assignments and project report successfully.

We wish her bright future.

For upSkill Campus

Program Coordinator

Date: 04/10/2026

For UCT

Project Mentor



Page 3 / 16 - ACKNOWLEDGEMENT

ACKNOWLEDGEMENT

I would like to express my sincere gratitude to all who helped me in completing this Industrial Internship Report.

First, I thank upSkill Campus, The IoT Academy and UCT for giving me this opportunity to work on real industrial problem statement.

I am thankful to my respected Principal and Faculty Members of Nabapravat Academy for Management Education for their continuous guidance.

I thank my parents and friends for their support.

I have learned Python, Data Cleaning, EDA, Machine Learning and Report Writing during this internship.

This report is made with full sincerity and effort.

Soumya Priyadarshinee Sahoo

Nabapravat Academy for Management Education

Date: 04-10-2026



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Page 5 / 16 - EXECUTIVE SUMMARY

EXECUTIVE SUMMARY

This report details the Industrial Internship provided by upSkill Campus and The IoT Academy in collaboration with UniConverge Technologies Pvt Ltd (UCT).

Internship was focused on project/problem statement provided by UCT. We had to finish project including report in 6 weeks.

My Project: Diabetes Prediction System using Machine Learning.

Objective: Build ML model to predict whether person has diabetes based on medical parameters like Glucose, BP, BMI, Age etc.

Tools Used: Python, Pandas, Numpy, Matplotlib, Scikit-learn, JupyterLab, Anaconda Navigator.

Dataset: Pima Indians Diabetes Dataset (Kaggle) - 768 rows, 8 features + 1 target.

Workflow: Data Collection -> Preprocessing -> EDA -> Train Test Split -> Model Training -> Evaluation.

Models Used: Logistic Regression, Random Forest, SVM, KNN. Best Accuracy: Random Forest 81.5%.

This internship gave me great exposure to industrial problems and enhanced my technical skills.

Keywords: Diabetes Prediction, Machine Learning, Random Forest, Healthcare AI



Page 6 / 16 - CHAPTER 1: INTRODUCTION

CHAPTER 1: INTRODUCTION

1.1 Data Science:

Data Science is interdisciplinary field using scientific methods to extract knowledge from data. It is most demanding field in 2026.

1.2 ML in Healthcare:

Healthcare generates huge data. ML helps in:

- Early disease prediction
- Personalized treatment
- Medical imaging
- Diabetes is chronic disease - early prediction saves lives.

1.3 Purpose of Internship:

- Industrial exposure
- Learn Python for Data Science
- Understand ML lifecycle
- Work on real problem
- Improve report writing

1.4 Scope:

Develop automated system to predict diabetes with high accuracy. Can be used as screening tool in hospitals, clinics, rural health centers.

1.5 Objective of Report:

This report explains step-by-step work done during 6 weeks internship by Soumya Priyadarshinee Sahoo at Nabapravat Academy.



Page 7 / 16 - ABOUT ORGANIZATION

CHAPTER: ABOUT ORGANIZATION

1. upSkill Campus (USC):

Career development platform providing industrial internship in collaboration with top companies. Training in AI, ML, IoT, Web Dev.

2. The IoT Academy:

Leading training academy in IoT and AI, provides hands-on training.

3. UniConverge Technologies Pvt Ltd (UCT):

UCT is leading IoT and AI solutions provider, Delhi. Established 2013. Provides industrial solutions in IoT Hardware, AI/ML, Embedded Systems, Automation.

UCT collaborated with USC for Industrial Internship 2024.

Internship Structure:

Week 1: Python Basics, Libraries (Pandas, Numpy)

Week 2: Data Cleaning, Handling Missing Values, EDA

Week 3: Machine Learning Algorithms (Logistic, RF, SVM)

Week 4: Project Development

Week 5: Model Evaluation, Graphs

Week 6: Report Writing, Submission, Viva

Mentor Support: Weekly live sessions, Doubt clearing, LMS access.



Page 8 / 16 - PROBLEM STATEMENT

CHAPTER 2: PROBLEM STATEMENT

2.1 Problem Statement (Given by UCT):

"Build a Machine Learning model to predict whether a person has diabetes or not based on diagnostic measures."

Diabetes is when blood sugar is too high due to lack of insulin. Manual diagnosis is time-consuming. Need automated system.

2.2 Dataset Description:

Source: Pima Indians Diabetes Database, Kaggle

Rows: 768, Columns: 9 (8 features + 1 target)

Features:

1. Pregnancies: Number of times pregnant
2. Glucose: Plasma glucose (mg/dL)
3. BloodPressure: Diastolic BP (mm Hg)
4. SkinThickness: Triceps skin fold (mm)
5. Insulin: 2-Hour serum insulin (mu U/ml)
6. BMI: Body mass index
7. DiabetesPedigreeFunction: Family history function
8. Age: Age in years
9. Outcome: 0=No Diabetes, 1=Diabetes

2.3 Objective:

- Analyze dataset
- Build ML model with >75% accuracy
- Compare algorithms
- Create prediction system



Page 9 / 16 - SYSTEM REQUIREMENTS

CHAPTER 3: SYSTEM REQUIREMENTS

3.1 Hardware:

- Processor: Intel i3 or above / AMD equivalent
- RAM: 4GB min, 8GB recommended
- Storage: 500MB free
- Internet for dataset download

3.2 Software:

- OS: Windows 10/11, Linux, macOS
- Language: Python 3.8+
- IDE: Anaconda Navigator, Jupyter Notebook, VS Code
- Browser: Chrome

3.3 Libraries Used:

1. Pandas: Data manipulation, DataFrame
2. Numpy: Numerical operations
3. Matplotlib: Visualization, Graphs
4. Seaborn: Statistical graphs, Heatmap
5. Scikit-learn: ML algorithms
 - `model_selection.train_test_split`
 - `preprocessing.StandardScaler`
 - `ensemble.RandomForestClassifier`
 - `linear_model.LogisticRegression`
 - `metrics.accuracy_score`, `confusion_matrix`

3.4 Custom Module (UCT Guideline):

File: `AI_Aug_26.py`

Contains functions: `load_data()`, `clean_data()`, `train_model()`

Import: `import AI_Aug_26`

Installation:

`pip install pandas numpy matplotlib seaborn scikit-learn jupyter`



Page 10 / 16 - METHODOLOGY

CHAPTER 3: METHODOLOGY & WORKFLOW

Flow Diagram:

Data Collection -> Preprocessing -> EDA -> Train-Test Split -> Model Training -> Evaluation -> Prediction

Step 1: Data Collection

Download diabetes.csv from Kaggle, load via `pd.read_csv()`.

Step 2: Data Preprocessing

- No null values, but many 0 values in Glucose, BP, SkinThickness, Insulin, BMI (not possible)
- Replace 0 with median: `df[col].replace(0, median)`
- Check `df.info()`, `df.describe()`
- Feature Scaling: `StandardScaler`, $z=(x-\text{mean})/\text{std}$

Step 3: EDA

- Outcome distribution: 500 No Diabetes (65.1%), 268 Diabetes (34.9%)
- Correlation: Glucose 0.47 with Outcome (highest)
- Histogram, Boxplot for each feature
- Heatmap using `seaborn.heatmap()`

Step 4: Train-Test Split

X = 8 features, y = Outcome

`train_test_split(X,y,test_size=0.20,random_state=42)`

Train: 614, Test: 154

Step 5: Model Training

4 models trained: Logistic Regression, Random Forest (100 trees), SVM, KNN(k=5)

Step 6: Evaluation

Accuracy, Confusion Matrix, Precision, Recall, F1



Page 11 / 16 - CODE IMPLEMENTATION**CHAPTER 3: CODE IMPLEMENTATION**

```
# Import Libraries
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler
from sklearn.ensemble import RandomForestClassifier
from sklearn.metrics import accuracy_score, confusion_matrix

# Load Data
df = pd.read_csv('diabetes.csv')
print(df.head())

# Preprocessing
cols_zero = ['Glucose', 'BloodPressure', 'SkinThickness', 'Insulin', 'BMI']
for col in cols_zero:
    df[col] = df[col].replace(0, df[col].median())

# Split
X = df.drop('Outcome', axis=1)
y = df['Outcome']
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=42)

# Scaling
scaler = StandardScaler()
X_train = scaler.fit_transform(X_train)
X_test = scaler.transform(X_test)

# Train Random Forest
model = RandomForestClassifier(n_estimators=100, random_state=42)
model.fit(X_train, y_train)

# Predict & Accuracy
y_pred = model.predict(X_test)
acc = accuracy_score(y_test, y_pred)
print(f"Accuracy: {acc*100:.2f}%") # 81.5%
print(confusion_matrix(y_test, y_pred))
```



Page 12 / 16 - RESULTS & ANALYSIS

CHAPTER 4: RESULTS & ANALYSIS

Model Comparison (Test Data 154 samples):

Model	Accuracy
Logistic Regression	76.2%
Random Forest	81.5% [BEST]
Support Vector Machine	77.8%
KNN	74.3%

Conclusion: Random Forest best with 81.5%.

Confusion Matrix - Random Forest:

	Predicted No	Predicted Yes
Actual No	86	13
Actual Yes	15	40

Correct: 126 / 154

Feature Importance (Random Forest):

1. Glucose - 24%
2. BMI - 18%
3. Age - 15%
4. DiabetesPedigreeFunction - 12%
5. Pregnancies - 10%
6. BloodPressure - 8%
7. Insulin - 7%
8. SkinThickness - 6%

Medical Validation: Glucose is primary indicator, matches medical knowledge.

Accuracy achieved is good for 768 samples dataset.



Page 13 / 16 - VISUALIZATIONS

CHAPTER 4: VISUALIZATIONS

Graph 1: Model Comparison Bar Chart

X=Models, Y=Accuracy. Random Forest tallest at 81.5%

[Paste model_accuracy.png]

Graph 2: Correlation Heatmap

Seaborn heatmap shows Glucose vs Outcome 0.47 correlation.

[Paste heatmap screenshot]

Graph 3: Outcome Pie Chart

No Diabetes 65.1% (500), Diabetes 34.9% (268) - Imbalanced

Graph 4: Glucose Distribution by Outcome

Histogram: Diabetic patients have higher glucose (mean 141 vs 109)

Graph 5: Age vs Glucose Scatter

Older + High Glucose = Higher diabetes risk

Graph 6: Boxplot of BMI by Outcome

Tools: Matplotlib, Seaborn

Key Insight: Glucose, BMI, Age are top 3 predictors. Early screening helps early detection.

Include Screenshots of Jupyter Notebook output, Code, Graphs.



Page 14 / 16 - CONCLUSION

CHAPTER 5: CONCLUSION

Project "Diabetes Prediction System using Machine Learning" completed successfully as part of Industrial Internship by upSkill Campus & UCT.

Achievements:

- Understood complete Data Science lifecycle
- Cleaned and preprocessed real medical dataset
- Performed EDA with visualizations
- Trained and compared 4 ML models
- Achieved 81.5% accuracy with Random Forest
- Created custom module AI_Aug_26.py as per UCT guidelines
- Prepared 16-page detailed report

Learning Outcomes:

1. Technical: Python, Pandas, Numpy, Matplotlib, Scikit-learn, JupyterLab
2. Professional: Report writing, Problem solving, Time management
3. Domain: Healthcare analytics, Diabetes
4. Industrial: Understood real problem statements, 6 weeks deadline

Application: Model can be used as preliminary screening tool in hospitals. Not replacement for medical diagnosis but helps early detection.

This internship was very valuable and increased my interest in Data Science. I will pursue advanced projects.

Thank you to UCT and upSkill Campus.

Soumya Priyadarshinee Sahoo
Nabapravat Academy for Management Education



Page 15 / 16 - FUTURE SCOPE

CHAPTER 5: FUTURE SCOPE

Improvements possible:

1. Web App Deployment:

Deploy model using Streamlit / Flask, user enters values, gets prediction, host on Heroku / AWS.

2. IoT Integration (UCT Speciality):

Connect IoT glucose sensor + fitness band, real-time monitoring, alert doctor if high risk.

3. Deep Learning:

Use ANN with hidden layers, TensorFlow/Keras, may get 85-90% accuracy.

4. Larger Dataset:

Use bigger datasets from Kaggle or hospital data, include lifestyle, diet features.

5. Mobile App:

Android app for rural areas, support Odia/Hindi language.

6. EHR Integration:

Integrate with Electronic Health Records, automatic risk assessment.

7. Explainable AI:

Use SHAP/LIME to explain why predicted diabetes, helps doctors trust.

With these, system can become complete healthcare solution for Odisha and India.

Potential for startup in health-tech domain.



Page 16 / 16 - REFERENCES

REFERENCES

1. Dataset:

Pima Indians Diabetes Database - Kaggle

<https://www.kaggle.com/datasets/uciml/pima-indians-diabetes-database>

Smith et al. ADAP algorithm for diabetes forecast.

2. Libraries:

Pandas: <https://pandas.pydata.org/docs/>

Numpy: <https://numpy.org/doc/>

Scikit-learn: <https://scikit-learn.org/stable/>

Matplotlib: <https://matplotlib.org/>

Seaborn: <https://seaborn.pydata.org/>

3. Organizations:

upSkill Campus: <https://www.upskillcampus.com/>

The IoT Academy: <https://www.theiotacademy.co/>

UCT: <https://www.uniconvergetechnologies.com/>

4. Books:

Hands-On ML with Scikit-Learn & TensorFlow - Aurélien Géron

GeeksforGeeks ML Tutorials

5. Papers:

Kavakiotis et al. 2017 - ML and Data Mining in Diabetes Research

American Diabetes Association Standards

Thank You!

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